

## CS571 Fall 2023: Web Technologies Exam 2 - Results



### Attempt 1 of 1

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Attempt Score **87.41 / 100 - 87.41 %**

Overall Grade (Highest Attempt) **87.41 / 100 - 87.41 %**

### High-Performance Websites Questions

#### Question 1

1.2 / 2 points

The Performance Golden Rule is that 80%-90% is spent on the front-end. Therefore you should start there because \_\_\_\_\_. Select all that apply.

-  ☒ A) ... it is simpler to optimize
-  ☐ B) ... it has proven to work
-  ☒ C) ... it has great potential for improvement
-  ☐ D) ... Wilfredo Pareto Principle says so
-   ☒ E) ... it is much cheaper than changing back end

## Question 2

2 / 2 points

[2 Points] What is the most important rule for improving web site performance? Select one answer.

- ☐ Minify JavaScript
- ✓ ☒ Compress components
- ☐ Cache every resource
- ☐ Add Expire headers
- ☐ Minify CSS

## Responsive Website Design (RWD) Questions

### Question 3

2 / 2 points

[2 points] Consider the following media queries:

```
<link rel="stylesheet" media="only screen and (min-width:320px) and (max-width: 760px)" href="css1.css">
```

```
<link rel="stylesheet" media="only screen and (min-width:761px) and (max-width: 1100px)" href="css2.css">
```

What different device sizes they fail to support? Select all that apply.

- ✓ ☒ Devices with large screens
- ✓ ☒ Devices with small screens

### Question 4

2 / 2 points

[2 points] What are properties of fluid grids? Select all that apply.

- ✓ ☒ Elements will adjust their widths and heights by the specified proportions to their parent container
- ✓ ☐ Have to adjust width manually for certain devices
- ✓ ☒ Defined using relative dimensions
- ✓ ☐ Define grids based on number of rows and columns

### Question 5

2 / 2 points

[2 Points] Switching between display `x.style.display = "block";` and `x.style.display = "none";` may be used in RWD.

- ✓ ☒ a) True
- ☐ b) False

### Question 6

2 / 2 points

[4 points] Select which concepts are the basis of Responsive Website Design. Select all that apply.

- ✓ ☒ fluid grids
- ✓ ☒ CSS3 media queries
- ✓ ☐ content reduction
- ✓ ☒ flexible images

## Cookies and Privacy Questions

### Question 7

1.714 / 2 points

[2 points] Which of the following are different types of cookies? Select all that apply.

- ✓ ☒ Server-side cookies
- ✓ ☐ Fourth-party cookies
- ✓ ☒ Persistent cookies
- ✗ ☐ Conversion-tracking cookies
- ✓ ☒ Non-secure cookies
- ✓ ☒ Non-secure cookies
- ✓ ☒ Client-side cookies

### Question 8

2 / 2 points

[2 points] Select all statements that are true for the **HttpOnly** attribute of cookies. Select all that apply.

- ✓ ☒ A cookie with HttpOnly attribute can only be accessed by the web server.
- ✓ ☒ The HttpOnly attribute tells the browser not to make it discoverable through document.cookie.
- ✓ ☐ A cookie with HttpOnly attribute can only be accessed through a secure connection.
- ✓ ☒ A cookie with HttpOnly attribute cannot be accessed via client-side scripting languages.
- ✓ ☒ A cookie with HttpOnly attribute cannot be stolen easily via cross-site scripting attacks.

### Question 9

1.667 / 2 points

[2 points] Select all functions that cookies can do. Select all that apply.

- ➔ ✓ ☒ Track user IP address.
- ➔ ✓ ☒ Track user visits.
- ➔ ✓ ☒ Store site information.
- ✗ ☒ Track user e-mail.
- ✓ ☐ Write data to disk.
- ✓ ☐ Install malware.

### 1St Half Course Questions

### Question 10

2 / 2 points

[2 Points] Consider the following example from w3schools.com.

```
<!DOCTYPE html>
<html>
<body>

<button
onclick="removeHandler()">Try
it</button>

<p id="demo"></p>

<script>
document.addEventListener("mousemove",
myFunction);

function myFunction() {

document.getElementById("demo").innerHTML
= Math.random();
}

function removeHandler() {

document.removeEventListener("mousemove",
myFunction);
}
</script>

</body>
</html>
```

What happens when one clicks the "Try it" button?

- ✓ ☒ Nothing
- ☐ A random number is displayed immediately
- ☐ A random number is displayed every time you move your mouse in this document.

## Question 11

1.333 / 2 points

[2 Points] What are true statements of HTML "anchors"? Select all that apply.

-  ☒ To create the target for an anchor with fragment, you must use the 'id' attribute
-  ☒ Fragment identifiers and related IDs must be exactly matched
-  ☒ Must be unique within a document

▼ [Hide question 11 feedback](#)

You can have as many anchors in an HTML document, as you wish.

## Question 12

0 / 2 points

[2 Points] Consider the following Python code:

```
b = "Hello, World!"  
print(b[-5:-2])
```

What gets printed?





-  ☒ Wor
- ☐ rld
-  ☐ orl

## Question 13

2 / 2 points

[2 Points] Consider the following example from w3schools.com.

```
<!DOCTYPE html>
<html>
<body>

<p id="demo"></p>

<script>
const text = '{"name":"John",
"age":"function() {return 30;}",
"city":"New York"}';
const obj = JSON.parse(text);
obj.age = eval("(" + obj.age +
")");
document.getElementById("demo").innerHTML
= obj.name + ", " + obj.age();
</script>

</body>
</html>
```

What string is displayed in the browser window?

- ☐ A) John, function() {return 30;}
- ☐ B) <a syntax error>
- ☒ C) John, 30

▼ [Hide question 13 feedback](#)

See [https://www.w3schools.com/js/tryit.asp?filename=tryjson\\_parse\\_function](https://www.w3schools.com/js/tryit.asp?filename=tryjson_parse_function)

## Question 14

1.5 / 2 points

[2 Points] Select all the JavaScript statements that are syntactically correct.

- ☒ ☒ `var multipleValues = [ ];`
- ☐ ☒ `var multipleValues = new Array();`
- ☒ ☒ `var multipleValues = Array();`
- ☒ ☒ `var multipleValues = Array(5);`

## Question 15

2 / 2 points

[2 Points] Below are lines extracted from the CSS style sheet of google.com.

```
.gb_T a {  
    border: 1px solid #e5e5e5;  
    -webkit-box-shadow: 0 1px 2px rgba(0, 0, 0, 0.1);  
    box-shadow: 0 1px 2px rgba(0, 0, 0, 0.1);  
    background: #fff;  
    cursor: -moz-grabbing;  
    cursor: -webkit-grabbing;  
    margin: -1px;  
    visibility: visible;  
    z-index: 1001  
}
```

Select any HTML example of usage of the CSS declaration above. Select all the snippet swwhere this CSS is triggered.

- ☒ ☒ `<div class="gb_T">  
 <a href=http://www.example.com>example </a>  
</div>`
- ☒ ☐ `<div class="gb_T">  
 <b>href=http://www.example.com>example</b>  
</div>`
- ☒ ☐ `<a href=http://www.example.com  
 class="gb_T">example</a>`
- ☒ ☒ `<span class="gb_T">  
 <a href=http://www.example.com>example </a>`



</span>

### Question 16

0 / 2 points

[2 Points] Where are proxy servers utilized?

- ➡ ☐ Both, at the client and server sides.
- ☐ At the client side only
- ✗ ☒ At the server side only

### REST Questions

### Question 17

1 / 1 point

[1 Point] In REST Web Services, it is best practice to use the slash "/" to represent a parent-child relationship.

- ✓ ☒ A) True
- ☐ B) False

### Question 18

1 / 1 point

[1 Point] In REST Web Services, select the best URI.

- ✓ ☒ A) <http://www.boeing.com/airplanes/747>
- ☐ B) <http://www.boeing.com/airplanes/747.html>

### Question 19

1 / 1 point

[1 Point] In REST Web Services, select the best URI.

- ☐ A) <http://www.parts-depot.com/parts?part-id=00345>
- ✓ ☒ B) <http://www.parts-depot.com/parts/00345>

**Question 20****1 / 1 point**

[1 Point] In REST Web Services, POST and PUT may be used for "updating" items.

- ✓ ☒ A) True  
☐ B) False

**Question 21****1 / 1 point**

[1 Point] In REST Web Services, plain text and CSV format may be used in formatting the responses.

- ✓ ☒ A) True  
☐ B) False

**Question 22****0 / 1 point**

[1 Point] In REST Web Services, it is best practice to use nouns in the logical URI, not verbs.

- ➡ ☐ A) True  
✗ ☒ B) False

**Question 23****2 / 2 points**

[2 points] For REST Web Services, select the statements that are TRUE. Select all that apply.

- ✓ ☐ REST services are a W3C recommendation  
✓ ☒ REST services are platform independent  
✓ ☒ REST services can use HTTPS  
✓ ☐ REST services are language dependent  
✓ ☒ REST services do not offer security

**Question 24****2 / 2 points**

[2 points] When comparing REST with other Web Services approaches, what statements are TRUE? Select all that apply.

- ✓ ☒ A) XML-RPC protocol and REST use HTTP as a transport.
- ✓ ☒ B) REST is popular because of its simplicity.
- ✓ ☒ C) SOAP APIs are very robust
- ✓ ☐ D) Simple Object Access Protocol is simpler than XML-RPC and JSON-RPC.

## jQuery Questions

### Question 25

2 / 2 points

[2 Points] jQuery is not needed when DOM manipulation is not used.

- ✓ ☒ a) True
- ☐ b) False

### Question 26

2 / 2 points

[2 points] Which of the following selectors are part of the Basic selectors category? Select all that apply.

- ✓ ☒ Multiple
- ✓ ☒ Element
- ✓ ☐ Form

### Question 27

2 / 2 points

[2 points] What is the JavaScript object added by jQuery? Select all that apply.

- ✓ ☒ \$
- ✓ ☒ jQuery

### Question 28

0 / 2 points

[2 points] What type of selector is the name="value" selector, as in the following example:

```
$( "input[value='Hot Fuzz'] " ).text( "Hot Fuzz" ) ;
```

- ➡ ☐ Attribute Equals selector
- ✗ ☒ Attribute Text selector
- ☐ Attribute Contains selector

### Question 29

2 / 2 points

[2 points] What type of selector is the ':contains()' selector, as in the following example:

```
$( "div:contains('John')" ).css( "text-decoration", "underline" );
```

Select one correct answer.

- ☐ Child Filter category
- ☐ Child Contains category
- ✓ ☒ Content Filter category

### Question 30

2 / 2 points

[2 points] Which of the following objects are abstracted by jQuery? Select all that apply.

- ✓ ☒ XMLHttpRequest
- ✓ ☒ JSON
- ✓ ☒ DOM

## Serverless Questions

### Question 31

2 / 2 points

[2 points] Select features of a "container." Select all that apply.

- ✓ ☒ made mainstream by Docker
- ✓ ☒ portable
- ✓ ☐ requires a VM

### Question 32

2 / 2 points

[2 Points] Consider the following code snippet.

```

exports.handler = async (event, context) => {
    let body;
    let statusCode = '200';
    const headers = {
        'Content-Type': 'application/json',
    };

    try {
        switch (event.httpMethod) {
            case 'DELETE':
                body = await
dynamo.delete(JSON.parse(event.body)).promise();
                break;
            case 'GET':
                body = await dynamo.scan({ TableName:
event.queryStringParameters.TableName }).promise();
                break;
            case 'POST':
                body = await
dynamo.put(JSON.parse(event.body)).promise();
                break;
            case 'PUT':
                body = await
dynamo.update(JSON.parse(event.body)).promise();
                break;
            default:
                throw new Error(`Unsupported method
"${event.httpMethod}"`);
        }
    } catch (err) {
        statusCode = '400';
        body = err.message;
    } finally {
        body = JSON.stringify(body);
    }
}

```

```
    return {  
        statusCode,  
        body,  
        headers,  
    };  
};
```

What is it?

- ✓ ☒ An AWS Lambda function
- ☐ A Google Cloud Function

### Question 33

2 / 2 points

[2 points] Which of the following are properties of a container like Docker? Select all that apply.

- ✓ ☒ Containers on same host share single Linux kernel
- ✓ ☐ Not portable
- ✓ ☒ Container images can be shared

### Question 34

2 / 2 points

[2 points] Select which of the following is true of AWS Lambda? Select all that apply.

- ✓ ☒ Adapts to rates of incoming requests
- ✓ ☒ Functions are unit of deployment
- ✓ ☐ Can be triggered by HTTP **only** and events

## HTML5 Questions

### Question 35

2 / 2 points

[2 Points] Consider the following code from W3Schools.

```
<!DOCTYPE html>  
<html>  
<body>
```

```
<canvas id="myCanvas" width="300" height="150" style="border:1px solid #d3d3d3;">
```

Your browser does not support the HTML5 canvas tag.</canvas>

```
<script>
```

```
var c = document.getElementById("myCanvas");
```

```
var ctx = c.getContext("2d");
```

```
ctx.fillRect(20, 20, 150, 100);
```

```
</script>
```

```
</body>
```

```
</html>
```

What gets displayed?

- ☐ A square filled with black
- ☐ A rectangle filled with red
- ✓ ☒ A rectangle filled with black
- ☐ A square filled with red

### Question 36

2 / 2 points

[2 points] Which of the following are removed elements in HTML5? Select all that apply.

- ✓ ☒ center
- ✓ ☐ embed
- ✓ ☒ font
- ✓ ☐ footer
- ✓ ☒ frameset

## JavaScript Frameworks Questions

### Question 37

2 / 2 points

[2 points] Which of the following are true of ReactJS? Select all that apply.

- ✓ ☒ Components are building blocks of React apps

✓ ☐ ReactJS can be used to build Android and iOS apps

Use RN instead

### Question 38

2 / 2 points

[2 points] Select which of the following is NOT true of NodeJS.

- ✓ ☐ handles networking
- ✓ ☐ is event driven
- ✓ ☒ uses blocking I/O model

### Question 39

2 / 2 points

[2 points] Which of the following are true of TypeScript? Select all that apply.

- ✓ ☒ Supports Optionals
- ✓ ☐ Supports MVC
- ✓ ☒ Includes support for classes and modules

### Question 40

2 / 2 points

[2 points] Select which of the following is true of React. Select all that apply.

- ✓ ☐ Is a framework like Angular
- ✓ ☒ Does not implement MVC

## Ajax Questions

### Question 41

2 / 2 points

[2 points] Consider the following object:

```
var xhr = new XMLHttpRequest();
```

Which values should be checked in the callback? Select all that apply.

- ✓ ☒ xhr.statusText is OK
- ✓ ☒ xhr.status is 404
- ✓ ☒ xhr.readyState is 4

### Question 42

2 / 2 points



[2 points] What are some of the characteristics of Ajax applications? Select all that apply.

- ✓ ☐ They have static icons
- ✓ ☒ They are single page
- ✓ ☒ They present live content
- ✓ ☒ They have visual effects

### Question 43

2 / 2 points

[2 points] Consider the following code from one of the Ajax example shown in class:

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN">
<html><head><title>My Third Ajax Script</title>
<script src="script03.js" type="text/javascript"
language="Javascript"></script></head>
<body><div id="pictureBar"> </div></body></html>
```

And this JavaScript:

```
window.onload = initAll;
var xhr = false;
function initAll() { same as previously except it calls getPix }
function getPix() { xhr.open("GET", "flickrfeed.xml", true);
xhr.onreadystatechange = XXXXX1;
xhr.send(null);setTimeout("getPix()",5 * 1000); }
function showPictures() {
    var tempDiv = document.createElement("div");
    var tempDiv2 = document.createElement("div");
    if (xhr.readyState == 4) {
        if (xhr.status == 200) {
            tempDiv.XXXXX2 = xhr.responseText;
            var allLinks = tempDiv.getElementsByTagName("a");
            for (var i=1; i<allLinks.length; i+=2) {
                tempDiv2.appendChild(allLinks[i].cloneNode(true)); }
            allLinks = tempDiv2.getElementsByTagName("a");
            var randomImg = Math.floor(Math.random() *
allLinks.length);
            document.getElementById("pictureBar").innerHTML =
allLinks[randomImg].innerHTML;
        } else { alert("There was a problem with the request " +
xhr.status); } } }
```

What should be the values of the two items listed as XXXXX1 and XXXXX2?

- ✓ ☒ showPictures and innerHTML
- ☐ callback and loadHTML

### Question 44

1 / 1 point

[1 Point] Fonts on a web page may be requested across domains using CORS.

- ✓ ☒ True  
☐ False

#### Question 45

1 / 1 point

[1 Point] Images do not have cross-domain issues.

- ✓ ☒ A) True  
☐ B) False

▼ [Hide question 45 feedback](#)

Images are considered "safe" and can be retrieved cross-domain

#### Question 46

1 / 1 point

[1 Point] Netscape 7 and Mozilla Firefox 1.0 were the first browsers to "clone" XMLHttpRequest functionality from Internet Explorer.

- ✓ ☒ A) True  
☐ B) False

#### Question 47

1 / 1 point

[1 Point] The XMLHttpRequest object can be used to upload a file.

- ✓ ☒ True  
☐ False

#### Question 48

1 / 1 point

[1 Point] The web browser displays a visual progress indicator while the XMLHttpRequest is being processed.

- ☐ True  
✓ ☒ False

#### Question 49

1 / 1 point

[1 Point] During AJAX open(), what does it mean to pass in <false> to the third parameter and why is it poor practice?

- ☐ A) False means the request will be executed asynchronously. It is bad because it breaks form validation
- ✓ ☒ B) False means the request will be executed synchronously. It is bad because a time-consuming synchronous request on the main thread will degrade UX.
- ☐ C) False means the request will be executed synchronously. It is bad because a time-consuming synchronous request on the main thread will crash the browser.
- ☐ D) False means the request will be executed asynchronously. It is bad because the flow of the application gets disrupted and the order of code execution becomes ambiguous

▼ [Hide question 49 feedback](#)

False == synchronous. Synchronous requests on the main thread can be easily disruptive to the user experience and should be avoided; in fact, many browsers have deprecated synchronous XHR support on the main thread entirely. Synchronous requests are permitted in Workers.

## Question 50

0 / 1 point

[1 Point] What is the major difference between jQuery AJAX and XMLHttpRequest (XHR)?

- ➡ ☐ A) There is no major difference aside from syntactical sugar

- ☐ B) jQuery is optimized for asynchronous use cases while XHR is not
- ☐ C) jQuery does not support all browsers while XHR does
- ☒ D) jQuery is Google's re-implementation of XHR from scratch for improved usability and efficiency

▼ [Hide question 50 feedback](#)

XMLHttpRequest is the raw browser object that jQuery wraps into a more usable and simplified form and cross browser consistent functionality. All jQuery functions use XMLHttpRequest object in the background, but provide additional functionality that you don't have to do yourself.

#### Question 51

1 / 1 point

[1 Point] Which of the following is not true about AJAX?

- ☐ A) AJAX uses XMLHttpRequest to communicate with the server
- ☐ B) JQuery includes the same functionality as AJAX
- ☒ C) AJAX must be performed in your remote script
- ☐ D) AJAX can be performed on submission of a form by adding the onsubmit parameter in the form tag

#### Question 52

1 / 1 point

[1 Point] What variable is available to us inside the xmlhttp.onreadystatechange callback in order to check the state of our request?

- ☐ A) this.currentState
- ☐ B) this.currentStatus
- ✓ ☒ C) this.readyState
- ☐ D) this.state

▼ [Hide question 52 feedback](#)

```
xhttp.onreadystatechange = function() {  
  if (this.readyState == 4 && this.status == 200) {  
    document.getElementById("demo").innerHTML =  
    this.responseText;  
  }  
};
```

### Question 53

1 / 1 point

[1 Point] Which of the following methods dispatches the request to the server when using AJAX?

- ☐ a) if (xhttp.responseText.trim().length > 0) { return false; } return true;
- ☐ b) open(method, url, async)
- ✓ ☒ c) send();
- ☐ d) new XMLHttpRequest();

**Question 54****0 / 1 point**

[1 Point] What is true of the fetch() API?

- ➡ ☐ A) Provides a more powerful and flexible feature set than XMLHttpRequest()
- ☐ B) It is identical to jQuery's .ajax()
- ☐ C) Unlike jQuery ajax(), cannot set CORS mode
- ✗ ☒ D) All of the above

**Web Security Questions****Question 55****1 / 1 point**

Combining VPN and TOR makes a user much more easily traceable.

- ☐ A) True
- ✓ ☒ B) False

▼ [Hide question 55 feedback](#)

Combining anonymizing VPN + Tor adds an extra layer of encryption and anonymity making it virtually impossible to trace you.

**Question 56****1 / 1 point**

[2 Points] A DDoS attack occurs when a system is overwhelmed with malicious electronic traffic.

- ✓ ☒ a) True
- ☐ b) False

### Question 57

1 / 1 point

[2 Points] The major reason to deprecating V2 Onion Service is exposure to attacks.

- ✓ ☒ a) True
- ☐ b) False

### Question 58

2 / 2 points

[2 points] Which types of hacks can cause damage to the user on the client side? Select all that apply.

- ✓ ☒ Log user keystrokes
- ✓ ☒ Stealing cookies
- ✓ ☐ Denial of Service
- ✓ ☒ Install malicious software
- ✓ ☐ Deface pages, altering content

### Question 59

2 / 2 points

[2 points] Which of the following are client-side attacks? Select all that apply.

- ✓ ☒ Clickjacking
- ✓ ☐ Insufficient Authentication
- ✓ ☒ Browser and Plugin vulnerability
- ✓ ☒ Cross-site scripting (XSS)
- ✓ ☐ Cross-Site Request Forgery (CSRF)

### Question 60

1 / 2 points

[2 Points] What are true statements of "Diceware" generated passwords? Check all that apply.

- ➡ ✓ ☒ Creates passwords that are extremely secure
- ➡ ✗ ☐ Creates passwords that are easy to memorize

✓ ☐ Creates password will low entropy

✗ ☒ Created by rolling a dice 5 times

Done